DSA Experiment 1

Title: String operations with and without pointers to arrays

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```
#include <stdio.h>
#include <string.h>
void copy(char input[], char output[]);
int palindrome(char input[]);
void reverse(char input[]);
void substring(char input[]);
int stringlength(char input[]);
int main()
{
   char str[10], output[10];
   int ch, c;

   printf("\nEnter the string: ");
   scanf("%s", str);

   do
   {
      printf("\n\n1) copy\n2) palindrome\n3) reverse\n4) substring\n5) Exit");
      printf("\nEnter your choice: ");
      scanf("%d", &ch);

      switch(ch)
      {
         case 1:
            copy(str, output);
            break;

         case 2:
            c = palindrome(str);
            if (c == 1)
               printf("\nThe given string is a Palindrome.");
            else
               printf("\nThe given string is not a Palindrome.");
            break;
```

```
            case 3:
                reverse(str);
                break;

            case 4:
                substring(str);
                break;

            case 5:
                printf("\nExiting program.");
                break;

            default:
                printf("\nInvalid choice! Please select 1-5.");
        }
    } while (ch != 5);

    return 0;
}

int palindrome(char name[])
{
    int i = 0, j = 0;
    while (name[j] != '\0')
        j++;
    j--;
    while (i < j)
    {
        if (name[i] != name[j])
            return 0;
        i++;
        j--;
    }
    return 1;
}
void copy(char input[], char output[])
{
    int i;
    for (i = 0; input[i] != '\0'; i++)
    {
        output[i] = input[i];
    }
    output[i] = '\0';
```

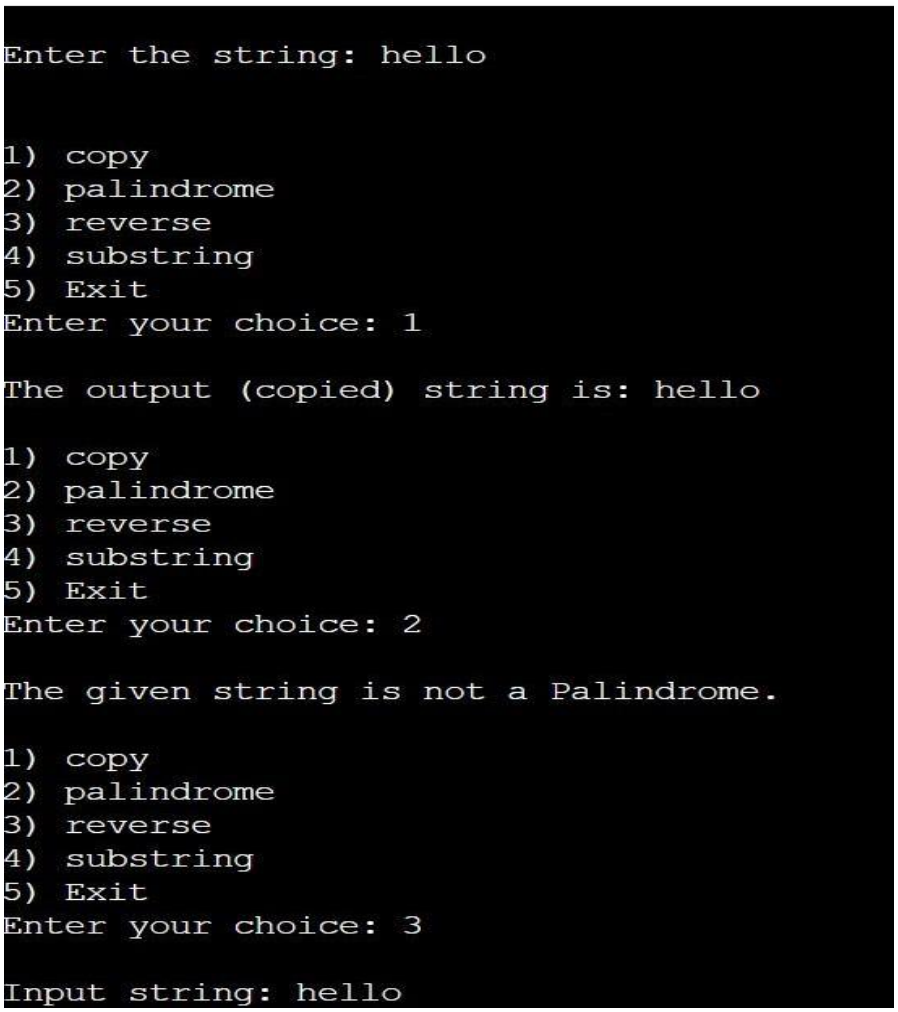
```
    printf("\nThe output (copied) string is: %s", output);
}

void substring(char input[])
{
    int i, n, position;
    char output[10];
    printf("\nEnter the starting position of substring (0-indexed): ");
    scanf("%d", &position);
    n = stringlength(input);
    if (position >= 0 && position < n)
    {
        for (i = 0; i < n - position; i++)
        {
            output[i] = input[position + i];
        }
        output[i] = '\0';
        printf("\nInput string: %s\nOutput substring: %s", input, output);
    }
    else
    {
        printf("\nPosition entered is out of range.");
    }
}

int stringlength(char input[])
{
    int i = 0;
    while (input[i] != '\0')
    {
        i++;
    }
    return i;
}
void reverse(char input[])
{
    int n, i;
    char output[10];
    n = stringlength(input);
    for (i = 0; i < n; i++)
    {
        output[n - 1 - i] = input[i];
    }
    output[n] = '\0';
```

```
    printf("\nInput string: %s\nOutput (reversed) string: %s", input, output);
}
```

OUTPUT:



```
Enter your choice: 3

Input string: hello
Output (reversed) string: olleh

1) copy
2) palindrome
3) reverse
4) substring
5) Exit
Enter your choice: 4

Enter the starting position of substring (0-indexed): 5

Position entered is out of range.

1) copy
2) palindrome
3) reverse
4) substring
5) Exit
Enter your choice: 4

Enter the starting position of substring (0-indexed): 2

Input string: hello
Output substring: llo
```

```
Input string: hello
Output substring: llo

1) copy
2) palindrome
3) reverse
4) substring
5) Exit
Enter your choice: 5

Exiting program.

...Program finished with exit code 0
Press ENTER to exit console.
```